

# Prasanjit Dubey (he/him)

Ph.D. Candidate, Machine Learning · Graduating May 2027

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| Contact Information | Email: <a href="mailto:dubey.prasanjit@gmail.com">dubey.prasanjit@gmail.com</a><br>Website: <a href="https://pdubey96.github.io">pdubey96.github.io</a><br>Google Scholar: <a href="#">Profile</a> | LinkedIn: <a href="https://in.linkedin.com/in/prasanjit-dubey">in/prasanjit-dubey</a><br>George Tower 1608<br>55 5th St NW, Atlanta, GA 30332 |
| Research Interests  | Multiple hypothesis testing, conformal prediction, anytime-valid inference, federated learning and knowledge distillation for language models, statistical learning theory.                        |                                                                                                                                               |

## Education

### Georgia Institute of Technology, Atlanta, GA, USA

**Ph.D. in Machine Learning** | GPA 3.85/4.00

Aug 2021 – May 2027 (expected)

- › H. Milton Stewart School of Industrial and Systems Engineering (ISyE), Minor in Mathematics
- › Advisor: Prof. Xiaoming Huo

### Indian Institute of Technology Bombay, Mumbai, India

**M.Sc. in Applied Statistics & Informatics** | Class Rank 1 of 38 | GPA 8.96/10.00

2017 – 2019

- › Department of Mathematics
- › Thesis: *Financial Data Mining Using Multivariate Copulas*

### University of Calcutta, Kolkata, India

**B.Sc. in Statistics**, First Class Honors | Grade 92.5%

2014 – 2017

- › Department of Statistics, Minor in Mathematics & Computer Science

## Publications and Preprints

Author order is by contribution. \* marks an award or a presentation associated with the paper.

### Manuscripts Under Review

- [1] **Report Resolution in Federated Multiple Testing under Family-Wise Error Control.**  
[Prasanjit Dubey](#) & Xiaoming Huo.  
*Under review at Biometrika, 2026.* [Paper]
- [2] **Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation under Heterogeneous Bandwidth Budgets.**  
[Prasanjit Dubey](#) & Xiaoming Huo.  
*Invited submission under review at the Journal of the American Statistical Association: Theory and Methods, 2026.* [Paper]  
\* Poster, Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X), Harvard University, 2026
- [3] **Anytime-Valid Federated Conformal RAG for LLM Swarms.**  
[Prasanjit Dubey](#) & Xiaoming Huo.  
*Under review at the Conference on Neural Information Processing Systems (NeurIPS), 2026.* [Paper]
- [4] **BOOST: Power-Optimal Strong-FWER Testing for Block-Structured Multiplicity.**  
[Prasanjit Dubey](#) & Xiaoming Huo.  
*Under review at the Conference on Neural Information Processing Systems (NeurIPS), 2026.* [Paper]
- [5] **Federated Language Models Under Bandwidth Budgets: Distillation Rates and Conformal Coverage.**  
[Prasanjit Dubey](#) & Xiaoming Huo.  
*Under review at the Conference on Neural Information Processing Systems (NeurIPS), 2026.* [Paper]
- [6] **Most Powerful Tests with Exact Family-Wise Error Rate Control: Necessary Conditions and Fast Computation for General K.**  
[Prasanjit Dubey](#) & Xiaoming Huo.  
*Under revision at the Journal of Computational and Graphical Statistics (JCGS), 2025.* [Paper]  
\* Invited talk, Optimisation for Statistics and Learning session, 2025 INFORMS Annual Meeting
- [7] **LLmFPCA-detect: LLM-powered Multivariate Functional PCA for Anomaly Detection in Sparse Longitudinal Texts.**  
[Prasanjit Dubey](#), Aritra Guha, Zhengyi Zhou, Qiong Wu, Xiaoming Huo, & Paromita Dubey.  
*Under review at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025.* [Paper]  
\* 2nd Place Paper Award, Student Research Showcase, 2025 INFORMS Regional Analytics Conference (IRAC)

## Refereed Publications

- [8] **Generalization Bounds for Magnitude-Based Pruning via Sparse Matrix Sketching.**  
Etash Kumar Guha, [Prasanjit Dubey](#), & Xiaoming Huo.  
*International Conference on Learning Representations (ICLR) Workshop on Bridging the Gap Between Practice and Theory in Deep Learning, 2024.* [Paper]

## Working Papers

- [9] **Pooling Sequential Evidence Across Hypotheses: Rate-Optimal Multiple Testing at a Fixed Horizon.**  
[Prasanjit Dubey](#) & Xiaoming Huo.  
*Manuscript in preparation, 2026.*
- [10] **Byzantine-Robust Federated RAG via Aligned Calibration and Fixed-Membership Conformal Prediction.**  
[Prasanjit Dubey](#), Aritra Guha, & Xiaoming Huo.  
*Manuscript in preparation, 2026.*

## Invited Talks & Presentations

- Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation**  
Poster, Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X), Harvard University 2026
- Most Powerful Test with Exact Family-Wise Error Rate Control: Necessary Conditions and a Path to Fast Computing**  
Invited talk, Optimisation for Statistics and Learning session, INFORMS Annual Meeting, Atlanta, GA 2025  
Poster, Georgia Statistics Day, University of Georgia, Athens, GA 2025
- LLmFPCA-detect: LLM-powered Multivariate Functional PCA for Anomaly Detection in Sparse Longitudinal Texts**  
Poster, Student Research Showcase, INFORMS Regional Analytics Conference (IRAC), Atlanta, GA 2025  
\* 2nd Place Paper Award

## Awards & Honors

- 2nd Place Paper Award**, Student Research Showcase, INFORMS Regional Analytics Conference (IRAC), Atlanta 2025
- Stewart Topper Fellowship Award**, Georgia Tech (first-year Ph.D. academic performance) 2021 – 2022
- Dr. P. V. Sukhatme Memorial Award**, IIT Bombay (M.Sc. rank 1) 2019
- All India Rank 26**, Graduate Aptitude Test in Engineering (GATE), Statistics 2019
- Pfizer Scholarship**, IIT Bombay (M.Sc. first-year rank 1) 2018
- All India Rank 7**, IIT Joint Admission Test for M.Sc. (IIT JAM), Mathematical Statistics 2017
- Swami Asaktananda Award**, University of Calcutta (highest B.Sc. academic distinction) 2017
- Summer Research Fellowship**, Indian Academy of Sciences 2016, 2017

## Research & Professional Experience

**Georgia Institute of Technology**, Atlanta, GA, USA Jan 2026 – Sep 2026  
*Graduate Research Assistant, Georgia Department of Transportation Sponsored Research*  
Data Sufficiency at Continuous Count Stations · PI: Prof. Xiaoming Huo [Code]

- Developed a **statistical framework** to establish **how many valid days a traffic-monitoring station needs** in a month with incomplete data before its monthly average can be published.
- Applied the framework to the entire network of **233 traffic-monitoring stations** across Georgia, delivering the results to transportation planners as the **Data-Sufficiency Explorer**, an interactive dashboard.

**Bloomberg L.P.**, New York, USA Jun 2025 – Aug 2025  
*Quant PhD Intern, Quantitative Risk Analytics*

- Built a **Kalman-filter hazard-curve model** for pricing credit instruments by estimating dynamic CDS term structures from **latent Nelson-Siegel factors and macroeconomic covariates**; **integrated the model into Bloomberg Terminal's** Credit VaR engine to forward-simulate risk-neutral hazard rates.
- Implemented a **Credit Rating Migration module** for the Credit VaR engine using **macro-conditioned multinomial logit and intensity-based models** to simulate obligor- and index-level rating paths, capturing systematic and idiosyncratic risk.

- › Developed a novel framework for analyzing **sparse longitudinal customer-journey text** by integrating **large language models (LLMs)** with **multivariate functional data analysis**; work resulted in a manuscript under review at **IEEE TPAMI** (paper [7]).
- › Used **LLM-guided dynamic keyword profiling** on real-world data (Amazon reviews) to surface customer pain points and actionable insights, **outperforming existing methods for customer-journey analytics**; shipped a **production application** (Emotional AI) for behavioral anomaly detection and targeted interventions.

Australia and New Zealand Banking Group (ANZ), Bengaluru, India  
Senior Quantitative Analyst, Wholesale Credit Risk Modeling

Jun 2019 – Aug 2021

- › Built the **Financial Ratio Surveillance Tool (FRST)**, a production **R Shiny** stress-testing application designed to forecast default probabilities and assess the predictive power of financial ratios.
- › Led the bank’s **statistical COVID-19 impact assessment** across commercial portfolios, flagging high-volatility sectors for stakeholders as part of **APRA regulatory** requirements.

Walmart Labs, Bengaluru, India  
Research Intern, Data Science (International Pricing & Demand Forecasting)

May 2018 – Aug 2018

- › Devised an **ensemble of ridge, lasso, and elastic-net models** using high-dimensional, large-scale data to forecast sales across product categories based on item-, region-, and covariate-specific **price elasticities**.
- › Built a **cluster-based (k-means) price-elasticity model**, **reducing mean absolute percentage error (MAPE) by 42.4% relative to Walmart’s incumbent methods**.

## Teaching Experience

Georgia Institute of Technology, Atlanta, GA, USA  
Head Teaching Assistant and Teaching Assistant, H. Milton Stewart School of Industrial and Systems Engineering

2021 – 2025

- › **ISyE 7406, Data Mining and Statistical Learning** (graduate level, 300+ students)  
Teaching Assistant: Fall 2024, Spring 2025, Fall 2025  
Instructor: Prof. Xiaoming Huo
- › **ISyE 6414, Statistical Modeling and Regression Analysis** (graduate level, 300+ students)  
**Head Teaching Assistant:** Fall 2022, Spring 2023 | **Teaching Assistant:** Summer 2022  
Instructor: Prof. Nicoleta Serban
- › **ISyE 2027, Probability with Applications** (undergraduate level)  
Student Assistant: Fall 2021 · Led tutoring sessions

## Student Mentoring

- › **Kareena Mehta**, undergraduate, Georgia Institute of Technology (Computer Science) Summer 2026
- › **Kevin Li**, undergraduate, Washington University in St. Louis (Finance and Data Science) Summer 2026, Fall 2026

## Professional Service

**Conference Reviewer:** International Conference on Learning Representations (ICLR) 2023 – present  
International Conference on Machine Learning (ICML)  
Conference on Neural Information Processing Systems (NeurIPS)

## Technical Skills

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|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming:</b>                  | Python, R, R Shiny, C++, SQL, MySQL, MATLAB                                                                                                                                                                                                           |
| <b>Machine Learning:</b>             | LLMs, PyTorch, scikit-learn                                                                                                                                                                                                                           |
| <b>Computing &amp; Tools:</b>        | High-performance computing (Slurm), GPU computing, Apache Hadoop (MapReduce, Streaming), Git, LaTeX                                                                                                                                                   |
| <b>Relevant Graduate Coursework:</b> | Mathematical Foundations of Machine Learning, High-Dimensional Probability, High-Dimensional Statistics, Statistical Inference, Time Series Analysis, Stochastic Processes, Convex Optimization, Stochastic Optimization, Computational Data Analysis |